VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
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CERTIFICATE

This is to certify that the Lab work entitled "Machine Learning (23CS6PCMAL)" carried out by **Dharunya Balavelavan (1BM23CS090),** who is a bonafide student of **B.M.S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

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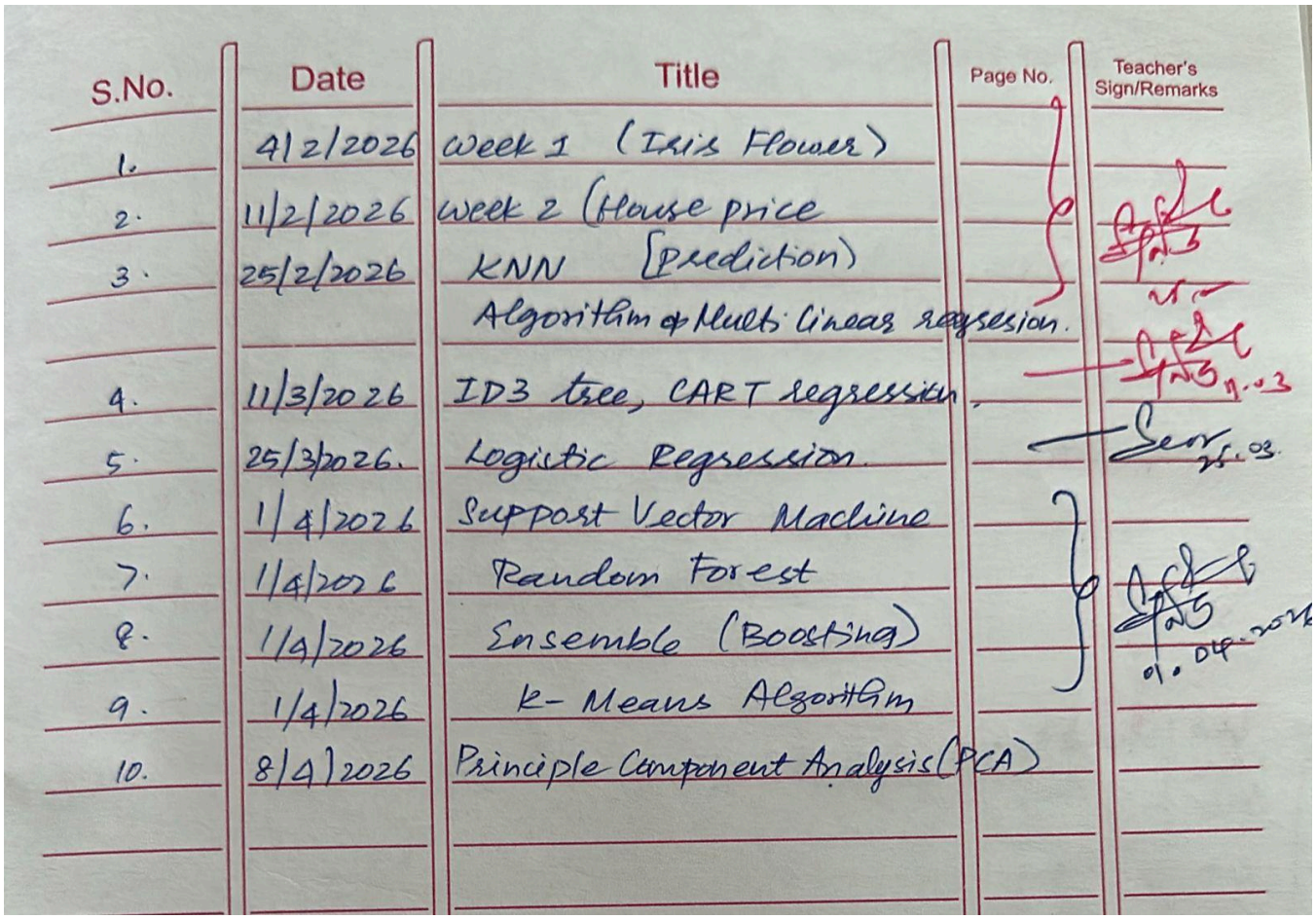
Github Link:

https://github.com/Dharunya21/MACHINE-LEARNING



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<u>Program 1</u>

Write a python program to import and export data using Pandas library functions

Screenshot

4/2/2026

Week-1

1). Tools:

a). Numpy: It is used for fast numerical operations and works with arrays.

b) Pandas: It is used to work with structured data (rows + column)

c) Matplotlib: It helps to visualize data using graphs and plots.

Task : 1 :

```
from sklearn.datasets import load_iris
import pandas as pd
iris = load_iris()
df = pd.DataFrame(iris.data, columns= iris.feature_names)
df['species'] = iris.target
df.head()
```

Index	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Columns = 5.

Task : 2

```
df.shape
```

=> (150, 5)

Features = 4.

Task : 3

df.columns

df.info()

#	column	Non-null count	Dtype
0	sepal length (cm)	150	float64
1	sepal width (cm)	150	float64
2	petal length (cm)	150	float64
3	petal width (cm)	150	float64
4	species (	150	int64

⇒ output column : species.

Task 4:

df.descsibe()

	sepal length	sepal width	petal length	petal width	species
count	150.000	150.0	150	150	150
mean	5.843	3.057	3.75	1.19	1
std	0.828	0.435	1.76	0.76	0.81
min	4.3	2.0	1.0	0.1	0
25%	5.1	2.8	1.6	0.3	0
50%	5.8	3.0	4.35	1.3	1
75%	6.4	3.3	5.1	1.8	2
max	7.9	4.4	6.9	2.5	2

⇒ feature with largest value = sepal length (≈ 7.9 cm)

Task : 5

df['species'].value_counts()

species	count
0	50
1	50
2	50

⇒ classes are balanced

Task : 6

import matplotl

plt.scatter(df['

plt.xlabel("se

plt.ylabel("

plt.show

⇒ Groups are

Task : 7

df[df['speci

df[df['spec

	sepal leng
50	7
51	6.4
52	6.9
53	5.5
54	6.5

⇒ Yes, val

specie

specie

Task : 8

df.c

s

sepal length

sepal width

petal length

petal width

species

Task : 6

```
import matplotlib.pyplot as plt
plt.scatter(df['sepal length'], df['petal length'])
plt.xlabel("sepal length")
plt.ylabel("petal length")
plt.show
```

⇒ Groups are formed.

Task : 7

```
df[df['species']==0].head()
df[df['species']==1].head()
```

	sepal length	sepal width	petal length	petal width	species
50	7	3.2	4.7	1.4	1
51	6.4	3.2	4.5	1.5	1
52	6.9	3.1	4.9	1.5	1
53	5.5	2.3	4.0	1.3	1
54	6.5	2.8	4.6	1.5	1

⇒ Yes, values do differ

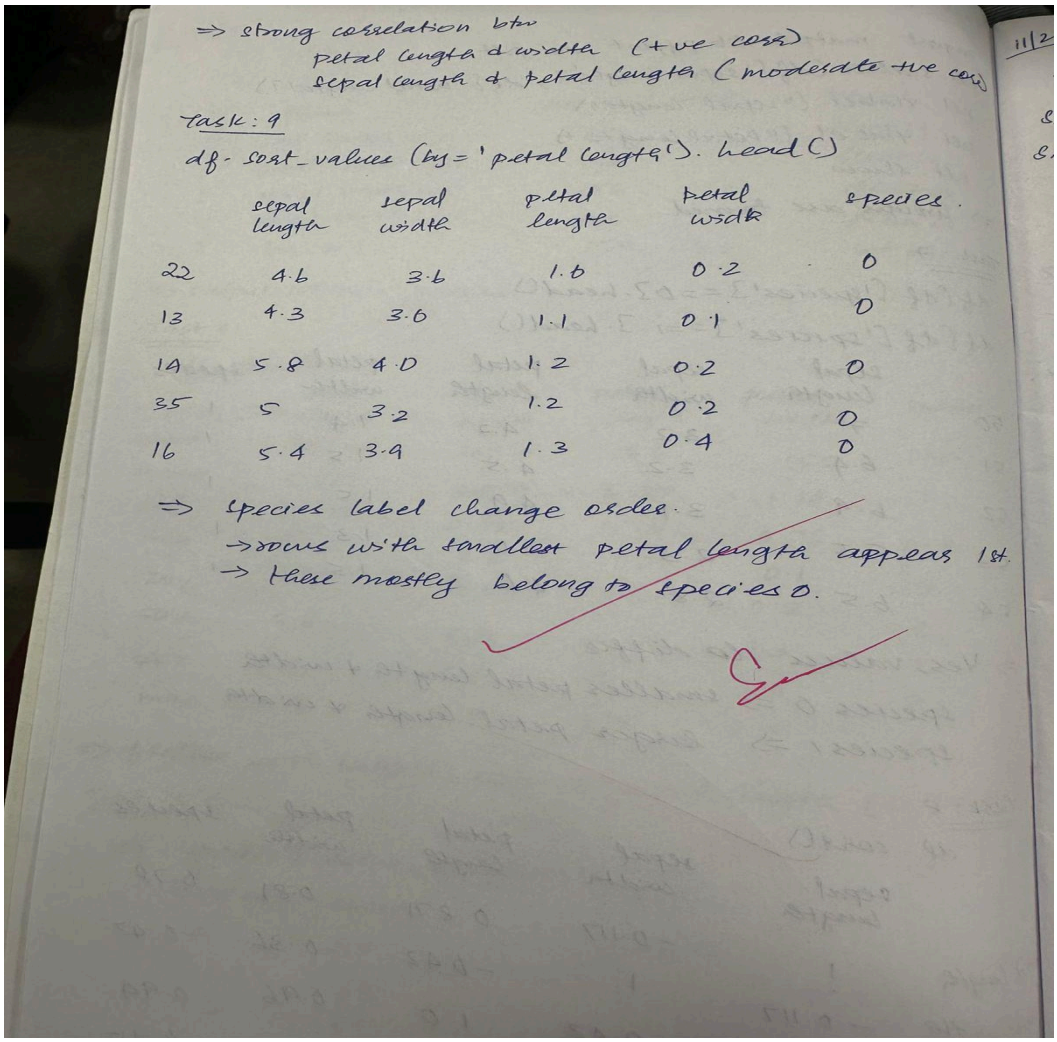
species 0 ⇒ smaller petal length & width

species 1 ⇒ larger petal length & width

Task : 8

```
df.corr()
```

	sepal length	sepal width	petal length	petal width	species
sepal length	1	−0.117	0.871	0.81	0.78
sepal width	−0.117	1	−0.42	−0.36	−0.42
petal length	0.817	−0.42	1.0	0.96	0.94
petal width	0.817	−0.36	0.96	1	0.95
species	0.782	−0.42	0.94	0.95	1



⇒ strong correlation btw
petal length & width (+ve corr)
sepal length & petal length (moderate +ve corr)

Task: 9

df.sort_values(by='petal length').head()

	sepal length	sepal width	petal length	petal width	species
22	4.6	3.6	1.0	0.2	0
13	4.3	3.0	1.1	0.1	0
14	5.8	4.0	1.2	0.2	0
35	5	3.2	1.2	0.2	0
16	5.4	3.9	1.3	0.4	0

⇒ species label change order.
→ rows with smallest petal length appears 1st.
→ these mostly belong to species 0.

Code:

```
from sklearn.datasets import load_iris

import pandas as pd

import matplotlib.pyplot as plt


# Load dataset

iris = load_iris()


# Create DataFrame
```

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target

# Basic info
print(df.head())
print(df.shape)
print(df.columns)
print(df.info())
print(df.describe())

# Class distribution
print(df['species'].value_counts())

# Scatter plot
plt.scatter(df['sepal length (cm)'], df['petal length (cm)'])
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.show()

# Filter data
print(df[df['species'] == 0].head())
print(df[df['species'] == 1].head())

# Correlation
```

```
print(df.corr())


# Sort values

print(df.sort_values(by='petal length (cm)').head())
```

Program 2

Demonstrate various data pre-processing techniques for a given dataset

Screenshot

11/2/2026.

Lab-2

House Price Prediction:

step 0 : Import required libraries.

Step 1 : Look at the big pictures.

problem type : supervised learning regression.

objectives : predict mean house value based on housing features.

evaluation metrics: RMSE

step 2 : get the data

step 3 : discover the data (EDA)

step 4 : visualize the data.

4.1 : distribution of house prices.

4.2 : income v/s house price.

4.3 : correlation heatmap

4.4 : location based visualization.

step 5 : Prepare the data

5.1 : feature engineering

5.2 : train-test

5.3 : feature scaling

step 6 : select and train Baseline Model.

6.1 Linear regression (0.74495, 0.53301, 0.57656)

step 7 : select & train advanced models.

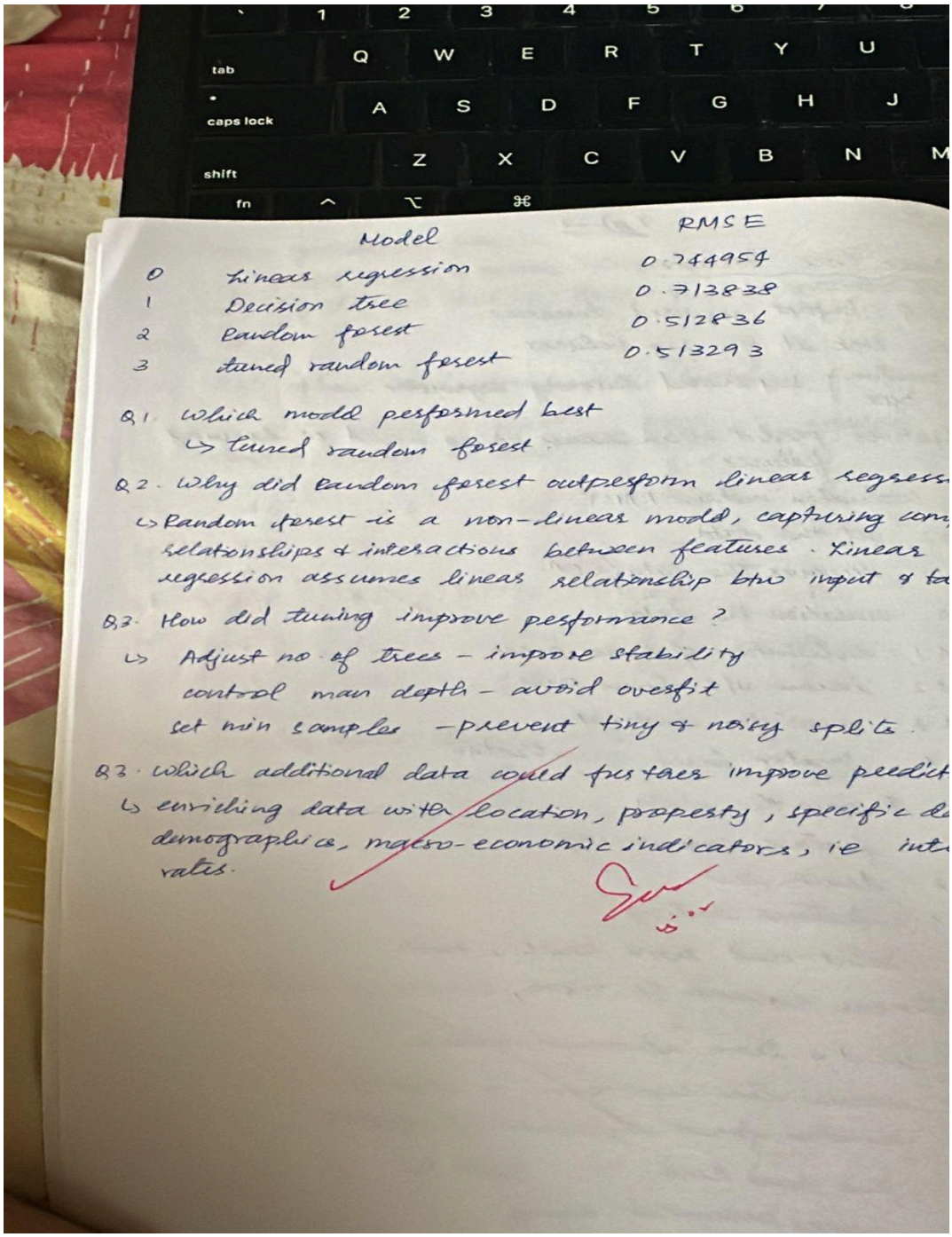
7.1 decision tree regression 0.7138773

7.2 random forest regression 0.51283 5.

Step 8 : Fine tune Model Best RM : 0.51329

8.1 hyper parameter tuning

step 9 : Model comparsion.



	Model	RMSE
0	linear regression	0.744954
1	Decision tree	0.713838
2	Random forest	0.512836
3	tuned random forest	0.513293

Q1. Which model performed best
↳ tuned random forest

Q2. Why did random forest outperform linear regress
↳ Random forest is a non-linear model, capturing com
relationships & interactions between features. Linear
regression assumes linear relationship btw input & ta

Q3. How did tuning improve performance?
↳ Adjust no of trees - improve stability
control max depth - avoid overfit
set min samples - prevent tiny & noisy splits.

Q3. Which additional data could features improve predict
↳ enriching data with location, property, specific de
demographics, macro-economic indicators, ie inte
rates.

Code:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
```

```
import seaborn as sns

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

# Load dataset
housing = fetch_california_housing(as_frame=True)

# Create DataFrame
df = housing.frame

# Display basic info
print(df.head())
print(df.shape)
print(df.info())

# Summary statistics
print(df.describe())
```

```
# Check missing values

print(df.isnull().sum())


# Distribution plot

plt.figure()

sns.histplot(df['MedHouseVal'], bins=30, kde=True)

plt.xlabel("Median House Value")

plt.ylabel("Frequency")

plt.title("Distribution of House Prices")

plt.show()


# Scatter plot

plt.figure()

sns.scatterplot(x=df['MedInc'], y=df['MedHouseVal'], alpha=0.5)

plt.xlabel("Median Income")

plt.ylabel("Median House Value")

plt.title("Income vs House Price")

plt.show()
```

Program 3

Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.

Screenshot

11/3/2026

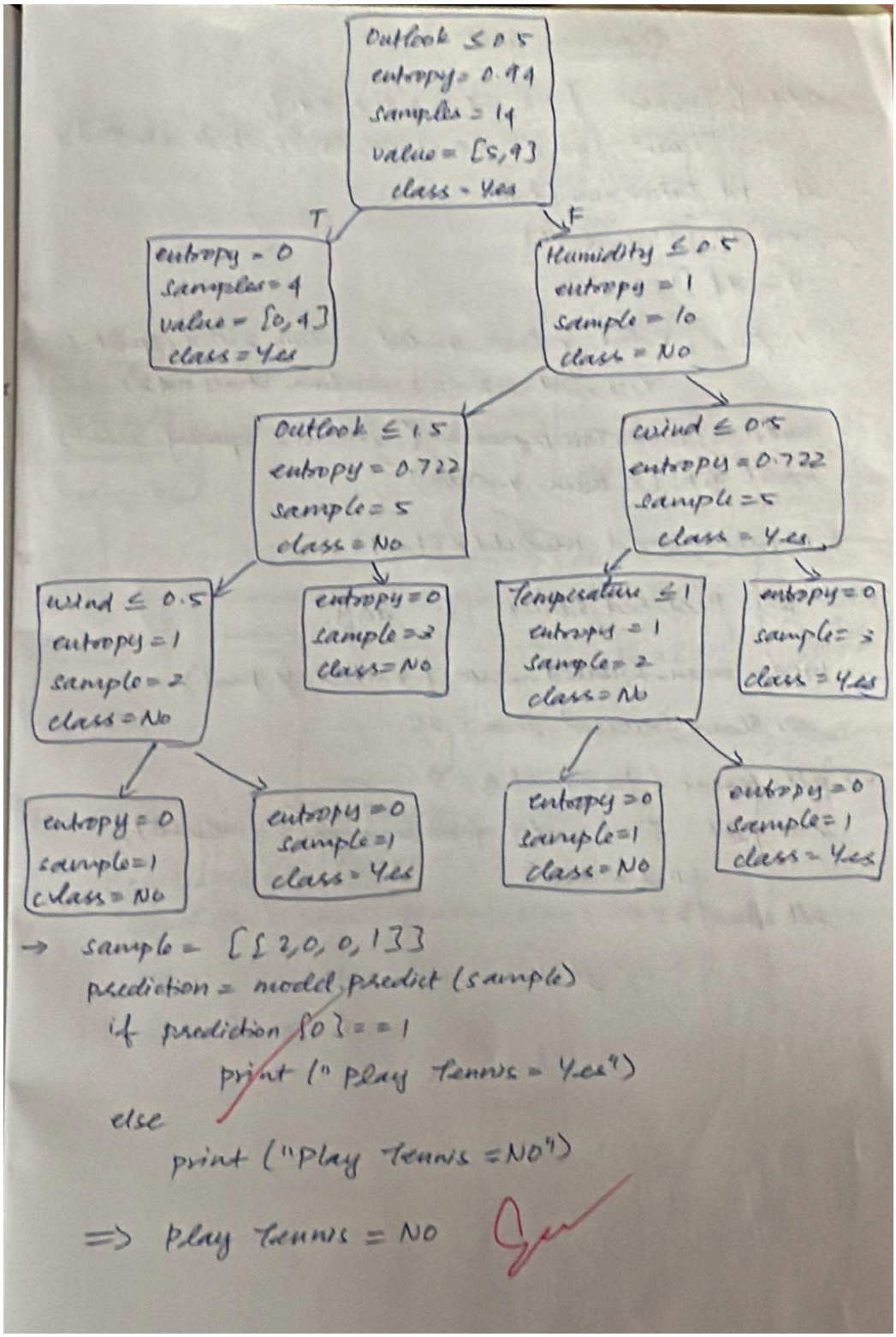
ID3 Tree

```
-> import pandas as pd
   from sklearn.tree import DecisionTreeClassifier
   from sklearn.preprocessing import LabelEncoder
   from sklearn import tree
   import matplotlib.pyplot as plt
-> df = pd.DataFrame(data)
```

=>

	Outlook	Temperature	Humidity	Wind	PlayTenn
0	Sunny	Hot	High	Weak	No
1	Overcast	Cool	Low	Strong	Yes

```
-> le = LabelEncoder()
   for column in df.columns:
       df[column] = le.fit_transform(df[column])
-> x = df.drop('PlayTennis', axis=1)
   y = df['PlayTennis']
-> model = DecisionTreeClassifier(criterion = "entropy")
   model.fit(x, y)
-> plt.figure(figsize = (12, 8))
   tree.plot_tree(model, feature_names = x.columns,
     class_names = ["No", "Yes"], filled=True)
   plt.show()
```



→ sample = [[2, 0, 0, 1]]

prediction = model.predict (sample)

if prediction [0] == 1

print ("Play Tennis = Yes")

else

print ("Play Tennis = NO")

⇒ Play Tennis = NO

Code:

import pandas as pd

```
from sklearn.tree import DecisionTreeClassifier

from sklearn.preprocessing import LabelEncoder

from sklearn import tree

import matplotlib.pyplot as plt



data = {

    'Outlook':
['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain','Sunny','Overcast','Ove
rcast','Rain'],

    'Temperature':
['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot','Mild'],

    'Humidity':
['High','High','High','High','Normal','Normal','Normal','High','Normal','Normal','Normal','High','Norm
al','High'],

    'Wind':
['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Weak','Weak','Strong','Strong','Weak','S
trong'],

    'PlayTennis': ['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes','No']

}



df = pd.DataFrame(data)

df



le = LabelEncoder()



for column in df.columns:

    df[column] = le.fit_transform(df[column])
```

```
df

X = df.drop('PlayTennis', axis=1)
y = df['PlayTennis']

model = DecisionTreeClassifier(criterion="entropy")
model.fit(X, y)

plt.figure(figsize=(12,8))
tree.plot_tree(model, feature_names=X.columns, class_names=["No","Yes"], filled=True)
plt.show()

sample = [[2,0,0,1]]   # example encoded values
prediction = model.predict(sample)

if prediction[0] == 1:
    print("Play Tennis: YES")
else:
    print("Play Tennis: NO")
```

Program 4

Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

Screenshot

Multi-Linear Regression.

Algorithm:

1. Collect data - gather features $x_1, x_2 \ldots x_n$ & target y.
2. Preprocess data
3. Define model - $y = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$
4. Choose cost function - usually mean squared error.
5. Optimize coeffxients - using normal equation.
6. Train model - fit the model & find β values.
7. Predict - use $\hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$
8. evaluate - check accuracy R^2, MSE, RMSE.

Output:

selected features: [bmi, bp, s1]

RMSE : 61.08

MAE : 49.99

R2 score : 0.29

coefficients : [39.53 18.96 0.46]

Intercept : 153.73

Code:

```
import numpy as np

from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score


# Load dataset

data = load_diabetes()

X = data.data
```

```
y = data.target

# Select features
selected_features = ['bmi', 'bp', 's1']

# Get feature indices
feature_names = data.feature_names
print("All features:", feature_names)

selected_indices = [feature_names.index(f) for f in selected_features]

# Extract selected features
X_selected = X[:, selected_indices]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_selected, y, test_size=0.2, random_state=42
)

# Standardization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Train Linear Regression model
lin_reg = LinearRegression()
lin_reg.fit(X_train_scaled, y_train)

# Predictions
y_pred_lr = lin_reg.predict(X_test_scaled)

# Evaluation
rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))
mae_lr = mean_absolute_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)

print("Selected features:", selected_features)
print("RMSE:", rmse_lr)
print("MAE:", mae_lr)
print("R2 Score:", r2_lr)
print("Coefficients:", lin_reg.coef_)
print("Intercept:", lin_reg.intercept_)
```

Program 5

Build Logistic Regression Model for a given dataset

Screenshot

25/3/2026.

Logistic Regression

```
→ df = pd.read_csv("HR_comma_sep.csv")
  print(df.head())
  print(df.info())
```

satisfaction level	last_evaluation	number_project	average_monthly_hours	time_spend_company

work_accident	left	promotion_last_5years	department

```
→ print(df['left'].value_counts())
  sns.countplot(x='left', data=df)
  plt.title("Employee Retention Count")
→ corr = df.corr(numeric_only=True)
  plt.figure(figsize=(10,6))
  sns.heatmap(corr, annot=True, cmap='coolwarm')
  plt.title("Correlation Heatmap")
→ salary_retention = df.groupby('salary')['left'].mean()
  salary_retention.plot(kind='bar')
  plt.ylabel("Probability of leaving")
→ dept_retention = df.groupby('Department')['left'].mean()
  dept_retention.plot(kind='bar', figsize=(10,5))
  plt.ylabel("Probability of leaving")
  plt.xticks(rotation=45)
→ df = pd.get_dummies(df, columns=['salary', 'Department'],
        drop_first=True)
→ x = df.drop('left', axis=1)
  y = df['left']
→ x_train, x_test, y_train, y_test = train_test_split(x, y,
        test_size=0.2, random_state=42)
→ y_pred = model.predict(x_test)
```

→ accuracy = accuracy_score(y_test, y_pred)
print (confusion_matrix (y_test, y_pred)
print (classification_report (y_test, y_pred))

⇒. Accuracy : 0.78233
Confusion Matrix : [[2117 177]
[476 2303]]

Classification report :

	Precision	recall	f1-score	support.
0	0.82	0.92	0.87	2294
1	0.57	0.33	0.41	706
accuracy.			0.78	3000
macro avg	0.69	0.62	0.64	3000
weighted avg	0.76	0.78	0.76	3000

Observation.

(i) Important variables affecting retention
* Satisfaction level
* Salary
* Average monthly hours
* Promotion
* time spent in company
* Department

(ii) Model Accuracy (80%)

↳ Yes, it is good
* model predicts most cases correctly
* suitable for a basic model.

25.03.2026

Code:

from google.colab import files

```
uploaded=files.upload()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("HR_comma_sep.csv")

# Preview
print(df.head())
print(df.info())

print(df.describe())

print(df['left'].value_counts())
sns.countplot(x='left', data=df)
plt.title("Employee Retention Count (0 = Stayed, 1 = Left)")
plt.show()

corr = df.corr(numeric_only=True)
plt.figure(figsize=(10,6))
sns.heatmap(corr, annot=True, cmap='coolwarm')
```

```
plt.title("Correlation Heatmap")
plt.show()

salary_retention = df.groupby('salary')['left'].mean()

salary_retention.plot(kind='bar')
plt.title("Impact of Salary on Employee Retention")
plt.ylabel("Probability of Leaving")
plt.show()

dept_retention = df.groupby('Department')['left'].mean()

dept_retention.plot(kind='bar', figsize=(10,5))
plt.title("Department vs Employee Retention")
plt.ylabel("Probability of Leaving")
plt.xticks(rotation=45)
plt.show()

# One-hot encoding
df = pd.get_dummies(df, columns=['salary', 'Department'], drop_first=True)

X = df.drop('left', axis=1)
y = df['left']
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Program 6

Build KNN Classification model for a given dataset.

Screenshot

5

28/02/2026

Lab-3

KNN (K-Nearest Neighbours) Algorithm:

* Choose a value for k:
  we first choose a value for K. This determines the no. of nearest neighbours used to make prediction.
* Calculate the distance:
  After choosing k, we calculate the distance between each data point in the training set and the target data point for which a prediction is being made.
* Find the k nearest neighbour:
  After calculating the distance between the existing data points and the new data point, we identify K nearest neighbours by selecting the K data points nearest to the new data point.
* Calculate the prediction:
  After finding the neighbours, we calculate the value of the dependent variable for the new data point. Then, we take the average of the target values of the K nearest neighbours.

Application:

California Housing Prices

```
=> data = fetch_california_housing()
   x = data.data
   y = data.target

=> x_train, x_test, y_train, y_test = train_test_split
      (x, y, test_size = 0.2, random_state = 42)

=> knn = KNeighboursRegressor (n_neighbours = 3)
   knn.fit (x_train, y_train)
```

Code:

```
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsRegressor

# Load dataset
data = fetch_california_housing()
X = data.data
y = data.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Standardization
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Train KNN Regressor
knn = KNeighborsRegressor(n_neighbors=3)
knn.fit(X_train, y_train)
```

```
# New data prediction
new_house = np.array([[8.3252, 41, 6.984, 1.023,
                       322, 2.555, 37.88, -122.23]])

new_house_scaled = scaler.transform(new_house)

predicted_price = knn.predict(new_house_scaled)

print("Predicted House Price (in $100,000s):", predicted_price[0])
print("Predicted Price in Dollars: $", predicted_price[0] * 100000)
```

Program 7

Build Support vector machine model for a given dataset.

Screenshot

11/9/2026.

Support Vector Machine

```
x = np.array([[1,60,50],[2,65,55] ... 3])
y = np.array([0,0,0,1,1,1,1,1,1,0,0,0,1,1,1,1,1,1])
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
model = SVC(kernel='linear')
model.fit(x_train, y_train)
```

```
=>  SVC
    SVC(kernel='linear')
```

```
y_pred = model.predict(x_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification report:", classification_report
    (y_test, y_pred))
```

=> Accuracy = 1.0

Classification report:

	precision	recall	f1-score	support
0	1	1	1	2
1	1	1	1	2

```
new_student = np.array([[5,70,65]])
new_student = scaler.transform(new_student)
prediction = model.predict(new_student)
if prediction[0] == 1:
    print("Result: Pass")
else:
    print("Result: fail")
```

=> Result: pass.

Code:

```
import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, classification_report


X = np.array([

    [1, 60, 50],

    [2, 65, 55],

    [3, 70, 60],

    [4, 75, 65],

    [5, 80, 70],

    [6, 85, 75],

    [7, 90, 80],

    [8, 95, 85],

    [2, 50, 45],

    [3, 55, 50],

    [4, 60, 55],

    [5, 65, 60],

    [6, 70, 65],

    [7, 75, 70],

    [8, 80, 75],
```

```
    [9, 85, 80]
])

y = np.array([
    0,0,0,1,1,1,1,1,
    0,0,0,1,1,1,1,1
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = SVC(kernel='linear')   # try 'rbf' also
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
new_student = np.array([[5, 70, 65]])  # example input
new_student = scaler.transform(new_student)


prediction = model.predict(new_student)


if prediction[0] == 1:
    print("\nResult: Pass")
else:
    print("\nResult: Fail")
```

Program 8

Implement Random forest ensemble method on a given dataset.

Screenshot

Random forest

```
df = pd.read_csv("random_forest_sample.csv")
df.head()
x = df.iloc[:, :-1]
y = df.iloc[:, -1]
x_train, x_test, y_train, y_test = train_test_split(
        x, y, test_size=0.2, random_state=42)
model = RandomForestClassifier(n_estimators=100,
        random_state=42)
y_pred = model.predict(x_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification report:", classification_report
        (y_test, y_pred))
print("Confusion Matrix:", confusion_matrix(y_test,
        y_pred))
```

=> Accuracy: 1.0

classification Report:

	precision	recall	f1-score	support
0	1	1	1	3
1	1	1	1	1

Confusion Matrix:

```
[[3 0]
 [0 1]]
```

Code:

```
import numpy as np

import pandas as pd
```

```
from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


# Load Iris dataset (optional reference)

from sklearn.datasets import load_iris

data = load_iris()

X = data.data

y = data.target


print("Feature shape:", X.shape)

print("Target shape:", y.shape)


# Create sample dataset

data = {

    "feature1": [5.1,4.9,4.7,4.6,5.0,5.4,6.4,6.9,5.5,6.5,5.7,6.3,6.5,7.6,4.9,7.3,6.7,7.2],

    "feature2": [3.5,3.0,3.2,3.1,3.6,3.9,3.2,3.1,2.3,2.8,2.8,3.3,3.0,3.0,2.5,2.9,2.5,3.6],

    "feature3": [1.4,1.4,1.3,1.5,1.4,1.7,4.5,4.9,4.0,4.6,4.5,4.7,5.8,6.6,4.5,6.3,5.8,6.1],

    "feature4": [0.2,0.2,0.2,0.2,0.2,0.4,1.5,1.5,1.3,1.5,1.3,1.6,2.2,2.1,1.7,1.8,1.8,2.5],

    "target":   [0,0,0,0,0,0,1,1,1,1,1,1,2,2,2,2,2,2]

}


df = pd.DataFrame(data)
```

```
# Save CSV
file_name = "random_forest_sample.csv"
df.to_csv(file_name, index=False)

# (Colab) Download file
from google.colab import files
files.download(file_name)

# (Colab) Upload file
uploaded = files.upload()

# Load dataset
df = pd.read_csv("random_forest_sample.csv")

print(df.head())
print(df.info())

# Split features and target
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
)

# Train Random Forest model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predictions
y_pred = model.predict(X_test)

# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Program 9

Implement Boosting ensemble method on a given dataset.

Screenshot

Boosting ensemble method.

```
data = load_breast_cancer (as_frame = True)
df = data.frame
df.head()
x = df.drop('target', axis=1)
y = df['target']
x_train, x_test, y_train, y_test = train_test_split (
        x, y, test_size=0.2, random_state=42)
model = AdaBoostClassifier (n_estimators = 50,
                random_state =42)
model.fit (x_train, y_train)
y_pred = model.predict (x_test)
accuracy = accuracy_score (y_test, y_pred)
print ("Accuracy")
cm = confusion_matrix (y_test, y_pred)
```

=> Accuracy = 0.9649

```
[[40, 3]
 [1, 70]]
```

Code:

import pandas as pd

import numpy as np

from sklearn.datasets import load_breast_cancer

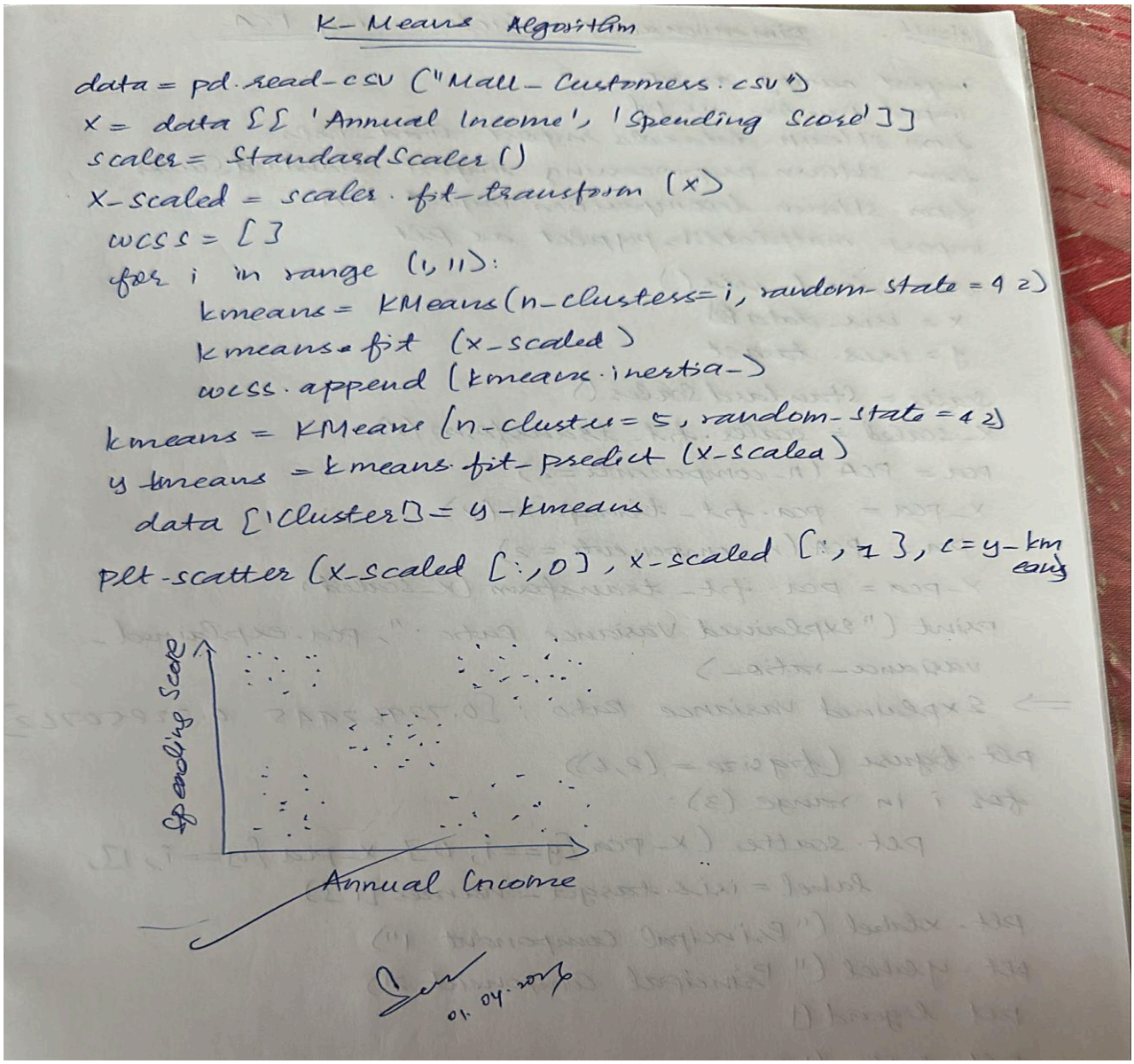
```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
data = load_breast_cancer(as_frame=True)
df = data.frame

# Features and target
X = df.drop('target', axis=1)
y = df['target']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train AdaBoost model
model = AdaBoostClassifier(n_estimators=50, random_state=42)
model.fit(X_train, y_train)
```

```
# Predictions
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```

Program 10

Build k-Means algorithm to cluster a set of data stored in a .CSV file.

Screenshot



K-Means Algorithm

```
data = pd.read_csv ("Mall_Customers.csv")
x = data [[ 'Annual Income', 'Spending Score']]
scaler = StandardScaler()
x_scaled = scaler.fit_transform (x)
wcss = [ ]
for i in range (1, 11):
    kmeans = KMeans(n_clusters= i, random_state = 42)
    kmeans.fit (x_scaled)
    wcss.append (kmeans.inertia_)
kmeans = KMeans (n_cluster= 5, random_state = 42)
y_kmeans = kmeans.fit_predict (x_scaled)
data ['Cluster'] = y_kmeans
plt.scatter (x_scaled [:,0], x_scaled [:, 1], c=y_kmeans
```


01.04.2026

Code:

```
from google.colab import files

uploaded = files.upload()


import pandas as pd

import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
```

```
from sklearn.preprocessing import StandardScaler

# Load dataset
data = pd.read_csv("Mall_Customers.csv")
print(data.head())

# Select features
X = data[['Annual Income (k$)', 'Spending Score (1-100)']]

# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Elbow method (WCSS)
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow Graph
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
```

```
plt.ylabel("WCSS")
plt.show()

# Apply KMeans (k=5)
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)

# Add cluster labels
data['Cluster'] = y_kmeans
print(data.head())

# Plot clusters
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_kmeans, cmap='rainbow')
plt.title("Customer Segments")
plt.xlabel("Annual Income (scaled)")
plt.ylabel("Spending Score (scaled)")
plt.show()
```

Program 11

Implement Dimensionality reduction using the Principle Component Analysis (PCA) method.

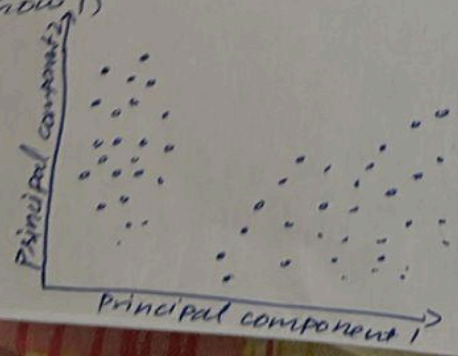
Screenshot

5/4/2026. **Dimensionality reduction using PCA.**

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA.
import matplotlib.pyplot as plt

iris = load_iris()
x = iris.data
y = iris.target
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
pca = PCA(n_components=2)
x_pca = pca.fit_transform(x)
pca = PCA(n_components=2)
x_pca = pca.fit_transform(x_scaled)
print("Explained Variance Ratio:", pca.explained_
variance_ratio_)
```

⟹ Explained Variance Ratio : [0.72962445 0.22850762]

```
plt.figure(figsize=(8,6))
for i in range(3):
    plt.scatter(x_pca[y==i, 0], x_pca[y==i, 1],
        label = iris.target_names[i])
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.title("PCA on Iris Dataset")
plt.show()
```



Code:

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
iris = load_iris()
X = iris.data
y = iris.target
# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
# Apply PCA (2 components)
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
# Explained variance
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
print("Total Variance Retained:", sum(pca.explained_variance_ratio_))

# PCA components (feature contributions)
print("PCA Components:\n", pca.components_)
```

```
# Plot
plt.figure(figsize=(8, 6))
for i in range(3):
    plt.scatter(X_pca[y == i, 0], X_pca[y == i, 1], label=iris.target_names[i])


plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.title("PCA on Iris Dataset")
plt.show()
```